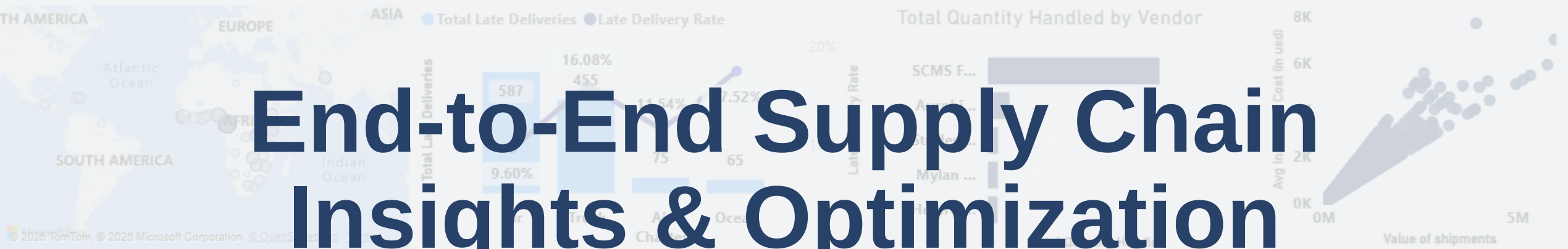


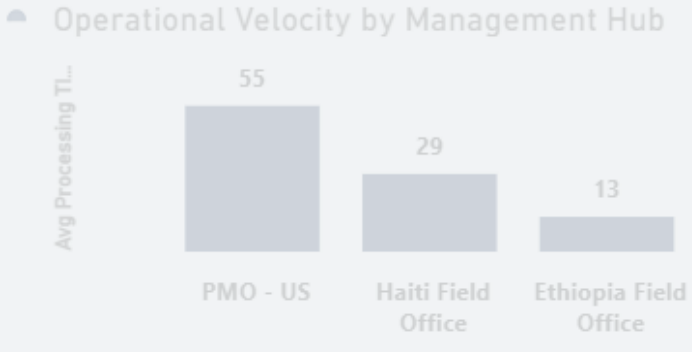
Global Freight Cost Distribution



End-to-End Supply Chain Insights & Optimization

Operational Performance, Risk Compliance, and Procurement Audit

Country	Vendor	Avg_freight_cost_per_kg
		56.38
		29.82
		25.77
		42.24
		132.01
		33.97



Processing Bottlenecks by Mode		
Management Hub	Shipment Mode	Avg Processing Time (in days)
Ethiopia Field Office	Truck	13
Haiti Field Office	Air	29
PMO - US	Air	30
PMO - US	Ocean	33
PMO - US	Air Charter	44
PMO - US	Truck	159

Logistics Audit

10.3K

Total Shipments

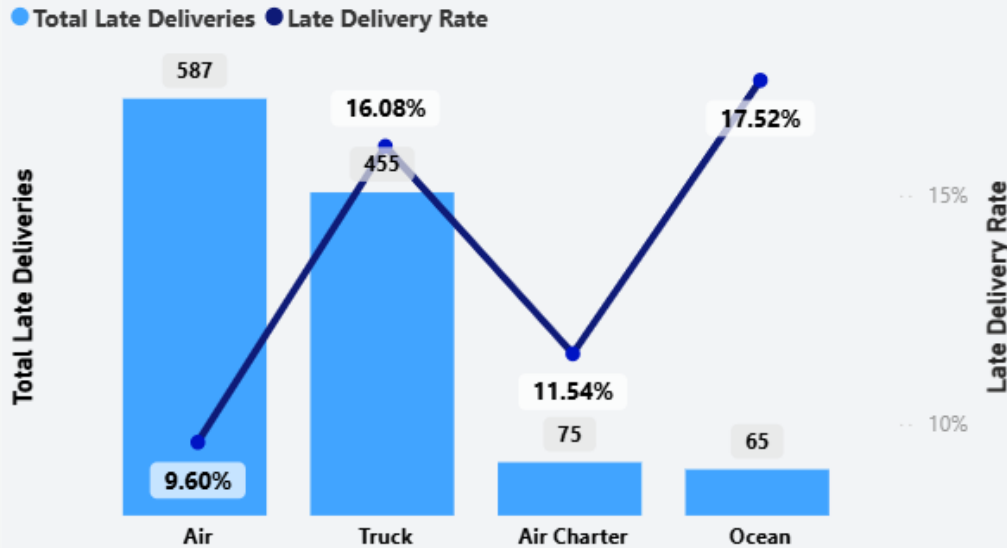


1.2K

Total Late Deliveries



- **Total Volume:**
10,300 shipments successfully audited across the global network.
- **Service Gap:**
Identified a baseline late delivery rate of 11.6% (1.2K units).
- **Observation:**
While Air has the highest raw count of late deliveries (587), Ocean has the highest late rate (17.52%).
- **Recommendation:**
If the delivery is time-sensitive, the data suggests Air is the most reliable (9.60%), while Ocean remains the least reliable mode with a 17.52% late rate.

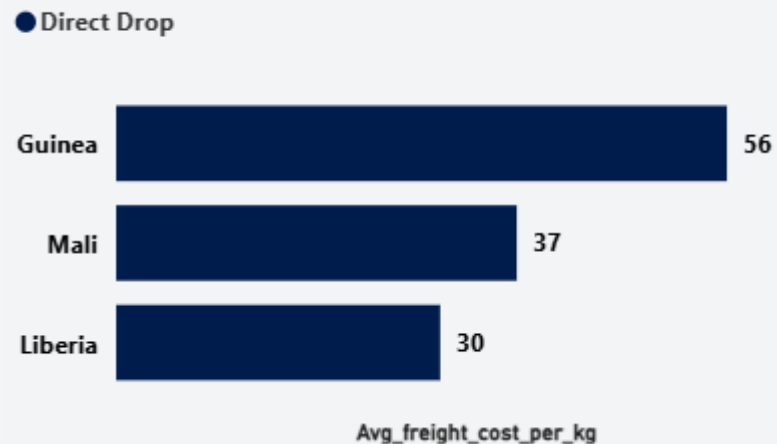


Regional Freight Cost Analysis

Global Freight Cost Distribution



Showing Top 3 countries (All shipped via Air & Direct Drop)



- **Observation:**
Freight costs are heavily skewed towards West African destinations when utilizing Air and Direct Drop modes.
- **Specific Insight:**
Guinea maintains the highest average freight cost per kg (56), significantly out-spacing other regional hubs, followed by Mali (37) and Liberia (30).
- **Recommendations:**
 1. For time-insensitive goods, negotiate a transition to Ocean Freight or Regional Trucking.
 2. Conduct a vendor-specific audit in these three countries to identify lower-cost shipment alternatives.

Vendor Efficiency & Cost Correlation

Country	Vendor	Avg_freight_cost_per_kg (in usd)
Liberia	Trinity Biotech, Plc	42.24
Liberia	Standard Diagnostics, Inc.	25.77
Guinea	Orgenics, Ltd	56.38
Liberia	Orgenics, Ltd	29.82
Mali	Orgenics, Ltd	33.97
Mali	Orasure Technologies Inc.	132.01



- Observations:**

1. There is a massive \$96/kg price gap between Orgenics, Ltd & Orasure Technologies Inc. (nearly 4 times) in Mali.
2. Standard Diagnostics, Inc. is currently the most cost-effective partner for Liberia.
3. There is an inverse relationship between the weights and freight cost: “As the weight in kg increases, average freight cost per kg almost always goes down”.

- Recommendations:**

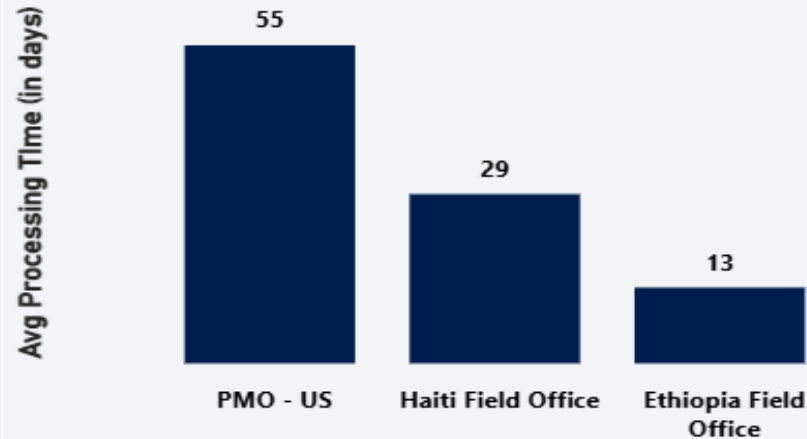
1. Audit Orasure Inc. contracts to identify cold chain requirements vs. vendor overcharging.
2. Implement batch shipping for low-weight cargo to capture 66%+ cost savings through consolidation.

Operational Velocity & Bottleneck Analysis

Processing Bottlenecks by Mode

Management Hub	Shipment Mode	Avg Processing Time (in days)
Ethiopia Field Office	Truck	13
Haiti Field Office	Air	29
PMO - US	Air	30
PMO - US	Ocean	33
PMO - US	Air Charter	44
PMO - US	Truck	159

Operational Velocity by Management Hub



- Observations:**

1. US Trucking Bottleneck: The PMO-US Management Hub is experiencing a severe bottleneck in its trucking operations, with a 159-day average processing time.

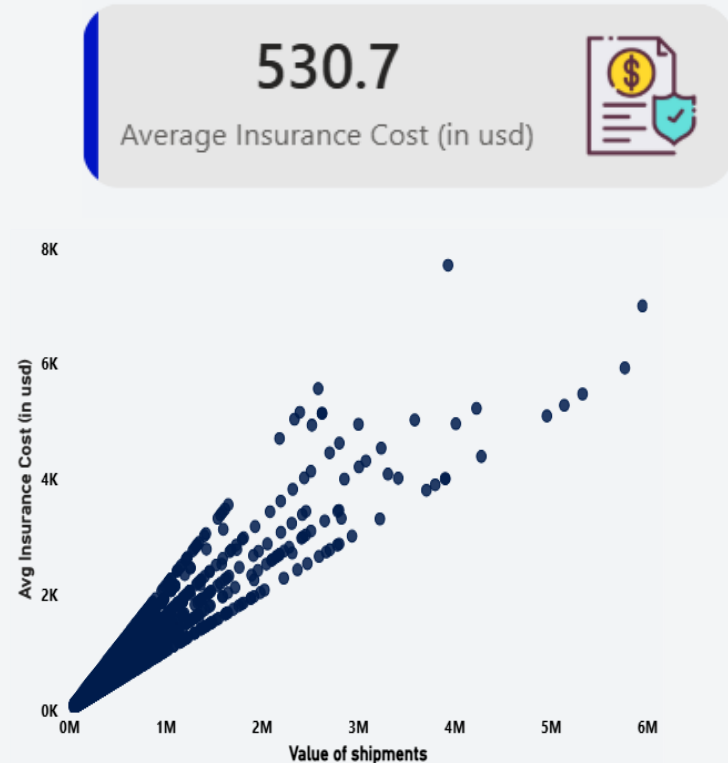
2. Comparative Efficiency: The Ethiopia Field Office is the most agile hub, processing shipments in just 13 days – over 12 times faster than the US trucking segment.

- Recommendations:**

1. Immediate Operational Review: Prioritize PMO-US lane audit to isolate customs, capacity, or carrier bottlenecks.

2. Knowledge Transfer: Scale Ethiopia's agile workflow to Haiti and US offices.

Risk Compliance & Insurance Value



- **Observations:**

1. The average insurance cost for shipments valued over \$50000 is \$530.7.
2. There are 113 high-value shipments (up to \$1.08M) currently in the system with no recorded insurance coverage.
3. Analysis confirms a linear relationship between cargo value and insurance premiums, indicating a standardized risk mitigation strategy.

- **Recommendations:**

1. Flag 113 uninsured IDs for Finance to verify third-party coverage or active exposure.
2. Mandate 'Hard Stop' in procurement for POs > \$50K with unverified insurance.

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Average Insurance Cost (in usd)



113.0

Total Shipments



Vendor Performance

Total Quantity Handled by Vendor



Total Quantity Handled

- **Observations:**
 1. **Market Dominance:** SCMS From RDC is the primary strategic partner, handling 135M units – which is more than 75% of the total volume handled by top 5 vendors combined.
 2. **Tier 2 Partners:** Aurobindo Pharma Limited (17M) and Strides Arcolab Limited (8M) serve as critical secondary supports, though their volume is significantly lower.
- **Recommendations:**
 1. **Risk Mitigation:** Establish contingency protocols for SCMS From RDC to ensure continuity during vendor strikes or failures.
 2. **Volume Leveraging:** Leverage 135M unit volume to negotiate superior freight rates and priority processing.

Integrated Supply Chain Control Center



“An end-to-end visualization suite enabling real-time monitoring of global logistics, vendor performance, and risk compliance.”

Project Impact & Strategic Summary

Operational Efficiency

Identified a 159-day lag in US Trucking; proposed a workflow audit based on the 13-day Ethiopia model.

Cost Optimization

Highlighted West Africa as a high-cost hub and flagged a \$96/kg price gap in Mali.

Risk & Compliance

Uncovered 113 high-value shipments without insurance; proposed a “Hard Stop” procurement protocol.

“This audit establishes a data-driven framework for future operational improvements and risk mitigation strategies.”

Thank You For Your Time!

Sanskriti Agrawal

“Data Analyst | Power BI & Risk Mitigation Specialist”



Connect for Collaboration

<https://www.linkedin.com/in/sanskriti-agrawal-506b79408/>



**View SQL Queries &
Documentation**

<https://github.com/SanskritiAgrawal123>

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